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# Michael Murray

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📍 San Francisco, CA

Portfolio & More: <https://linktr.ee/michaeljohnmurray>



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## PROFESSIONAL EXPERIENCE

Seasoned Mechanical Engineer with 15+ years of experience in **design, analysis, and manufacturing** across medical devices, power products, and automation. Demonstrated success in **product development, DFM (Design for Manufacturing), FEA (Finite Element Analysis), and cross-functional leadership**. Adept at **guiding teams** to deliver innovative, cost-effective solutions that comply with standards such as ANSI, ISO, FDA, IEC.

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### Nevados Engineering

Oakland, CA

#### Senior Mechanical Engineer

May 2025 – Present

- Redesigned the primary structural coupling on a solar tracker, moving to a four-bolt clamp that improved stiffness and more than doubled crack initiation life in accelerated testing.
- Ran FEA in SOLIDWORKS Simulation on couplings and torque tubes under downforce, uplift, and torsion to size geometry, welds, and material – validated against physical load and fatigue testing.
- Designed for manufacturability high-volume stamped and welded parts, cutting weld size and material cost without strength loss, and resolving tolerance and coating conflicts before tooling.
- Qualified gearmotor supplier, led failure analysis on a sheared output shaft, redesigned to prevent failure.
- Managed supplier quality and PPAP approvals, driving corrective action across overseas manufacturing.

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### Swope Design Solutions

San Francisco, CA

#### Senior Mechanical Engineer

June 2022 – May 2025

- **Led** a project to design and build a pump-to-catheter system for removing arterial calcified lesions.
- **Created** prototypes to establish proof of concept through CNC machining, molding, and 3D printing.
- **Developed** a positioning mechanism for a non-invasive surgical procedure to treat kidney stones.
- **Built** a camera-based automated data matrix scanning and analysis system (designed, wired, and coded)
- **Spearheaded** heater system design for a medical device, conducting thermal analysis and testing.
- **Managed** injection-molded part design and tooling, machined molds, and molded parts in silicone.
- **Redesigned** endoscopic robot components for injection molding (mass production), incorporating DFM.

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### True Digital Surgery

Santa Barbara, CA

#### Senior Staff Mechanical Design Engineer

November 2020 – November 2021

- **Designed** mechanical subsystems of a positioning system for a robotic neurosurgical 3D microscope.
- **Analyzed** and prevented mechanical failures and confirmed factor of safety per IEC 60601 requirements.
- **Performed** thermal analysis to maintain mechanical integrity under heat application.
- **Developed** fit/function checks using GD&T in accordance with ASME Y14.5 standards.
- **Co-authored** QCIC system structure per ISO 13485 guidelines.

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### Surround Medical Systems

Morrisville, NC

#### Senior Mechanical Engineer

May 2019 – May 2020

- **Led** the mechanical department to design a 3D x-ray positioning system from concept through production.
- **Improved** delivery of x-rays to reduce radiation exposure to patients and ensure image quality.
- **Co-authored** a 510(k) submission for FDA approval.
- **Prevented** mechanical failures through Finite Element Analysis (FEA) and Failure Mode and Effects Analysis (FMEA) studies.

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### ABB

Raleigh, NC

#### Senior Mechanical Engineer, R&D

July 2017 – September 2018

- **Standardized** transformer designs using common parts, reducing cost and lead times.
- **Enhanced** mechanical designs of transformers/enclosures to increase convection efficiency.
- **Developed** an outdoor-rated transformer cabinet incorporating weatherproof coating.

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### Ingersoll Rand

Charlotte Area, NC

#### Design Engineer III

August 2014 – July 2017

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- **Designed** parts/assemblies for complex, custom industrial centrifugal air compressors (2,000–7,000 cfm, 400–1,250 HP) to meet ANSI and ISO standards.
- **Designed** dual oil cooler system for 5000-cfm air compressors.
- **Optimized** airflow through pre-build assessments, improving compressor performance.

#### Newell Rubbermaid – Piedmont Hardware Brands

Charlotte Area, NC

#### Mechanical Design Engineer

February 2013 – August 2014

- **Designed** high-volume sheet metal, die cast, injection-molded, and stamped parts (metal and plastic).
- **Utilized** hand calcs and SOLIDWORKS Simulation for Finite Element Analysis, guiding new product designs.
- **Prevented** failure modes through Computer-Aided Engineering (CAE).

#### Sandvik

Mebane, NC

#### Manufacturing Engineer

September 2011 – February 2013

- **Generated** CAD models, CNC programs through CAM, and drawings for high-volume metal-cutting tools.
- **Reduced tooling costs** by 30% via optimized CAM programs for 5-axis milling machines, lathes, and grinders.
- **Designed** and implemented a coolant reclamation system to optimize machining efficiency and cut waste.

#### ABB

Lake Mary, FL

#### Mechanical Design Engineer

August 2009 – September 2011

- **Invented** a remotely operated device for industrial medium-voltage circuit breakers (5–15 kV); recognized with two patents (**U.S. Patents 9,876,335 and 8,654,513B2**).
- **Developed** a Modular Mechanism Operated Cell to retrofit circuit breakers (**Patent 8,878,088**).
- **Improved** circuit breaker thermal flow to prevent overheating and brazing issues.

### TECHNICAL SKILLS

|                                                                                                                                                                                                                                |              |        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------|
| <b>DFM, DFA, DFx</b> , (Design for Manufacturing and Assembly)                                                                                                                                                                 | Advanced     | 13 yrs |
| <b>CAD</b> , (Computer-Aided Design), Drafting<br><i>SOLIDWORKS, Creo Parametric, (Pro/E), NX, Autodesk Inventor, Onshape, Fusion</i>                                                                                          | Advanced     | 14 yrs |
| Product Development, Project Management, Jira, Confluence, and Others                                                                                                                                                          | Advanced     | 10 yrs |
| <b>FEA</b> , (Finite Element Analysis)                                                                                                                                                                                         | Advanced     | 9 yrs  |
| <b>GD&amp;T</b> , (Geometric Dimensioning & Tolerancing), Drafting, <b>ASME Y14.5</b>                                                                                                                                          | Advanced     | 9 yrs  |
| <b>FMEA</b> , Failure Mode and Effects Analysis)                                                                                                                                                                               | Advanced     | 7 yrs  |
| <b>PDM, PLM</b> , Software and Administration, <i>SOLIDWORKS PDM, Arena, Windchill</i>                                                                                                                                         | Advanced     | 6 yrs  |
| <b>Rapid Prototyping</b><br>• <b>3D Printing, Programming</b> : FDM, SLA, DMLS, SLS, Maintenance & Improvement of Machines<br>• <i>Assembly and use of tools and equipment to make and modify parts (full list on website)</i> | Advanced     | 10 yrs |
| Vendor Management                                                                                                                                                                                                              | Advanced     | 8 yrs  |
| CAM, (Computer-Aided Manufacturing), CNC Programming, CNC Operation                                                                                                                                                            | Intermediate | 4 yrs  |
| Thermal Analysis                                                                                                                                                                                                               | Intermediate | 4 yrs  |
| Computer-Aided Engineering ( <i>SOLIDWORKS Simulation, Flow</i> )                                                                                                                                                              | Intermediate | 4 yrs  |
| CAM, (Computer-Aided Manufacturing), ( <i>HSMWorks, NX</i> )                                                                                                                                                                   | Intermediate | 4 yrs  |
| Regulations and Standards: <b>FDA, IEC</b> , IEC 60601, ISO 13485, <b>ASME, ANSI, ISO, UL</b>                                                                                                                                  | Intermediate | 4 yrs  |
| Injection Molding, Plastics                                                                                                                                                                                                    | Intermediate | 5 yrs  |

### EDUCATION

**North Carolina State University, Mechanical Engineering, 2009**

**Honors:** Cum Laude, Dean's List, 2008 NCSU ME Design Award, Co-Op Program

### ADDITIONAL NOTES

**References:** Available upon request.

**Portfolio, Manufacturing Process Experience, and more:** <https://linktr.ee/michaeljohnmurray>